

BRAIN VR

A Senior Honors Thesis

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Introduction and Motivation: Bridging Neuroanatomy and Immersive Learning - Transforming Brain Education through a Virtual Reality Implementation of the Brain

The study of the human brain stands at the forefront of scientific inquiry, underpinning our understanding of thought, emotion, behavior, and the very essence of what it means to be human. Mastery of neuroanatomy is not only fundamental to medical and psychological sciences but is also increasingly relevant across a wide range of disciplines, from artificial intelligence to education and philosophy. However, the inherent complexity and three-dimensional structure of the brain make it one of the most challenging subjects to teach and learn.

For generations, traditional educational tools-such as textbooks, 2D illustrations, and static anatomical models-have been the primary resources for exploring the brain's intricate architecture. While these methods have laid the groundwork for foundational knowledge, they often fall short in capturing the brain's true spatial relationships, dynamic interconnections, and functional intricacies. Students are frequently left to imagine how structures fit together, interact, and contribute to the broader workings of the mind, resulting in gaps in comprehension and retention. Belonging to a family of doctors, I was really intrigued by the magnificence of the brain, but always wanted to learn the ins and outs of the organ in "real life". My curiosity to learn about the brain is what motivated me to work on creating an interactive VR model of the brain to make it easier to learn about the brain, and understand how each component of the brain contributes to the overall complex functionality of the brain.

The advent of virtual reality (VR) technology marks a transformative moment in the way we approach neuroanatomy education. VR offers an immersive, interactive, and highly visual

environment where learners can explore the brain in three dimensions, manipulate its components, and observe how different regions relate to one another in real time. This approach not only enhances spatial understanding but also fosters deeper engagement and curiosity, making complex concepts more accessible and memorable.

By harnessing the power of VR, educators and students can move beyond the limitations of traditional methods, opening new possibilities for experiential learning and interdisciplinary exploration. Teaching the anatomy of the brain through VR represents a significant leap forward- a futuristic, advanced, and more effective way to unlock the mysteries of our most vital organ. This thesis explores the development and educational potential of a VR-based brain model, highlighting its advantages and transformative impact on how we learn about the brain.

For my Senior Thesis, I developed a VR Model of the Human Brain in a Lab Setting, which allows users to explore the brain in real life. The brain consists of 141 interactive components, you can view information on the brain - the structural hierarchy tree, functions the selected component is responsible for, and what other components are associated with those functions. You can also enter an “exploded view” of the brain to make it easier to dive deeper into the anatomy.

Methods

1. Brain Mesh File Acquisition and Setup:

Firstly, the neuroanatomical dataset was sourced from the Allen Institute for Brain Science at <https://community.brain-map.org/t/human-brain-atlas-mesh-files/560>, comprising high-resolution annotation and segmentation files of the human brain. The annotation file was downloaded and loaded as a main image into itk-SNAP, an open-source software for medical image segmentation

and visualization, to convert the 3D volumetric data into OBJ mesh files. Next, the segmentation file was downloaded and loaded into itk-SNAP. Finally, the meshes were derived from the segmentation and exported as surface mesh, then imported into Unity. This process preserved the spatial relationships and anatomical accuracy of 141 distinct brain structures, which were later color-coded to enhance visual differentiation (Figure 1).

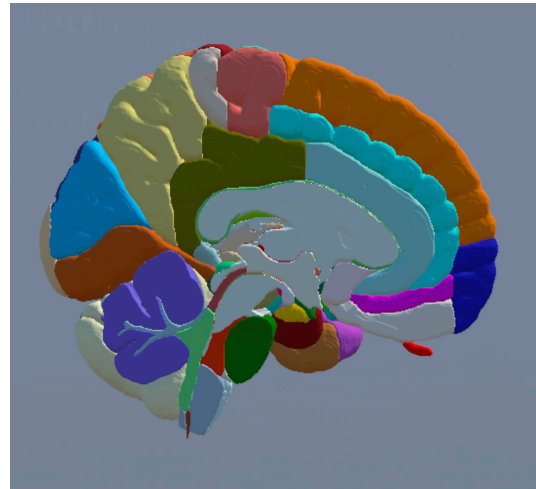


Figure 1

2. Unity Integration and Basic Modeling:

Each of the 141 mesh files were then assigned as separate child objects under a parent GameObject representing the whole brain. These components were assigned as mesh colliders to make it easier for VR interactions like Raycasting and grab interactions, and implementing the explosion-effect view since almost all of these components are irregular shaped.

3. ExplodeAndReturn:

Next, was the explosion-effect view section of the project. The exploded view functionality was implemented through a custom Unity script, ExplodeAndReturn, designed to toggle between intact and spatially separated brain configurations while synchronizing instructional interfaces. During initialization, the script captures and stores the original positions and rotations of all 141 brain components. Rigidbody components are programmatically added to each structure to enable physics-based interactions, ensuring that users can manipulate individual parts with realistic physical behavior. Figure 2 shows an exploded view of the brain.

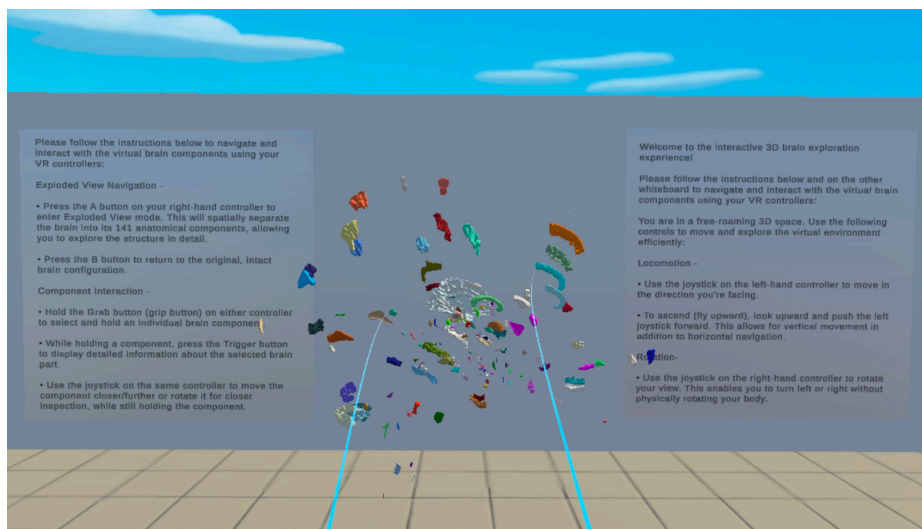


Figure 2

The Explode function calculates displacement vectors using each component's mesh centroid (mesh.bounds.center), scaled by a factor of 5 to achieve collision-free separation while preserving spatial relationships. This approach balances anatomical clarity with usability, allowing learners to inspect individual structures without losing contextual awareness. A state flag (exploded) prevents redundant transformations during transitions.

The Return function restores components to their original transforms with submillimeter precision, nullifying residual velocities to eliminate physics artifacts. Concurrently, instructional panels revert to default text using TextMeshPro (TMP), ensuring consistent guidance across interaction modes. The Unity Input System maps controller inputs to explosion/reset actions through an event-driven architecture, providing responsive, frame-accurate triggering of these features.

This implementation directly supports pedagogical objectives: the centroid-based displacement reinforces spatial learning theories by emphasizing structural relationships, while physics-

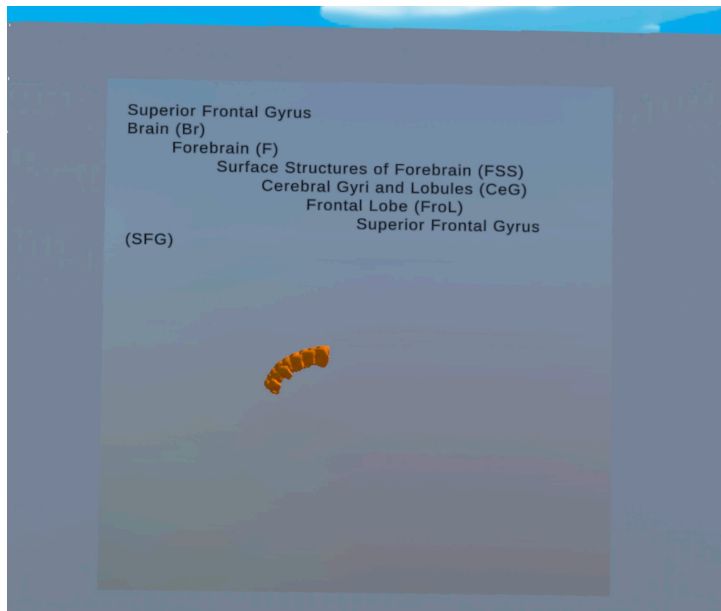


Figure 3

enabled manipulation fosters active exploration. The synchronized UI updates reduce cognitive load, allowing users to focus on either free exploration or guided instruction.

4. Brain Piece Information and Functions:

Next, the `BrainComponentInfo` and `BrainComponentFunc`

implementations were implemented to

display the hierarchical information and functions along with other components that are associated with the functions for each brain piece. This information would be displayed on two whiteboards - one for hierarchical information and the other for functions. These whiteboards are essentially Quads, and each of them have a prefab - `infoPanelPrefab` and `functionPanelPrefab`. Each of these prefabs then have a `TMPPro` (`TextMesh Pro`) unit attached to display the text (Figures 3 & 4).

`BrainComponentInfo` and `BrainComponentFunc` declare a reference to the info panel prefab, as well as public fields for the brain component's name and description (for `BrainComponentInfo`), and their

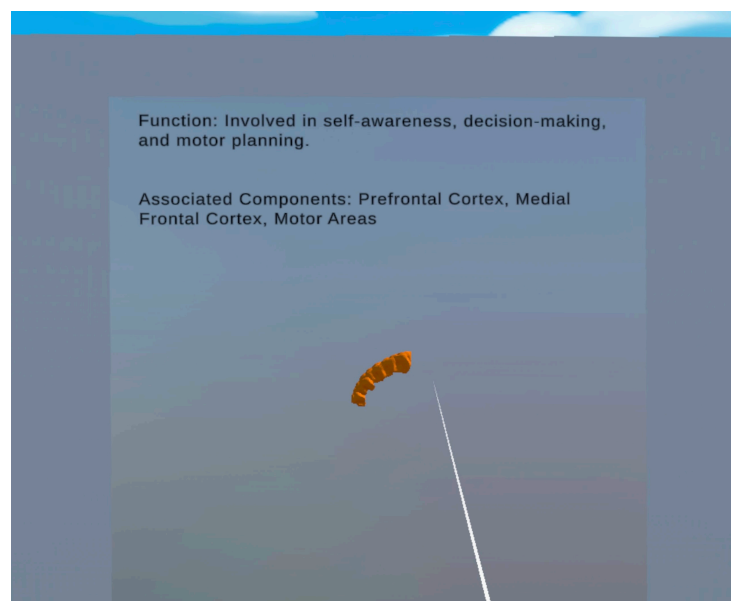


Figure 4

function and other components associated with those functions (for BrainComponentFunc) which can be set in the Unity Inspector. On startup, the script locates the TextMeshPro text component within the info panel. When the ShowInfo method is called-typically when a user

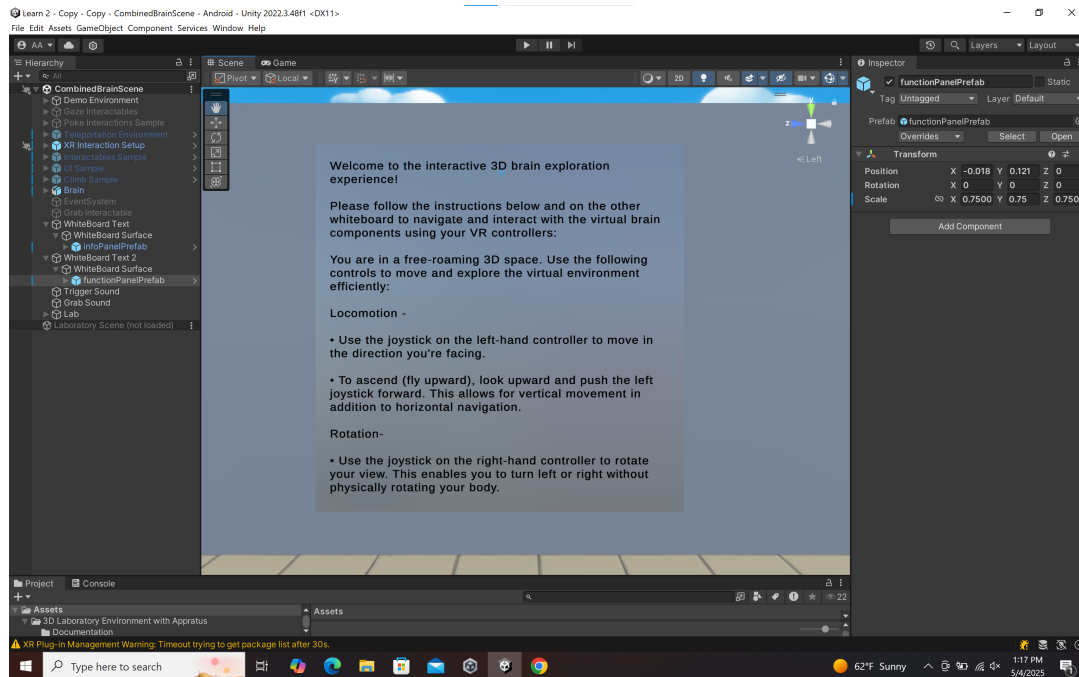


Figure 5

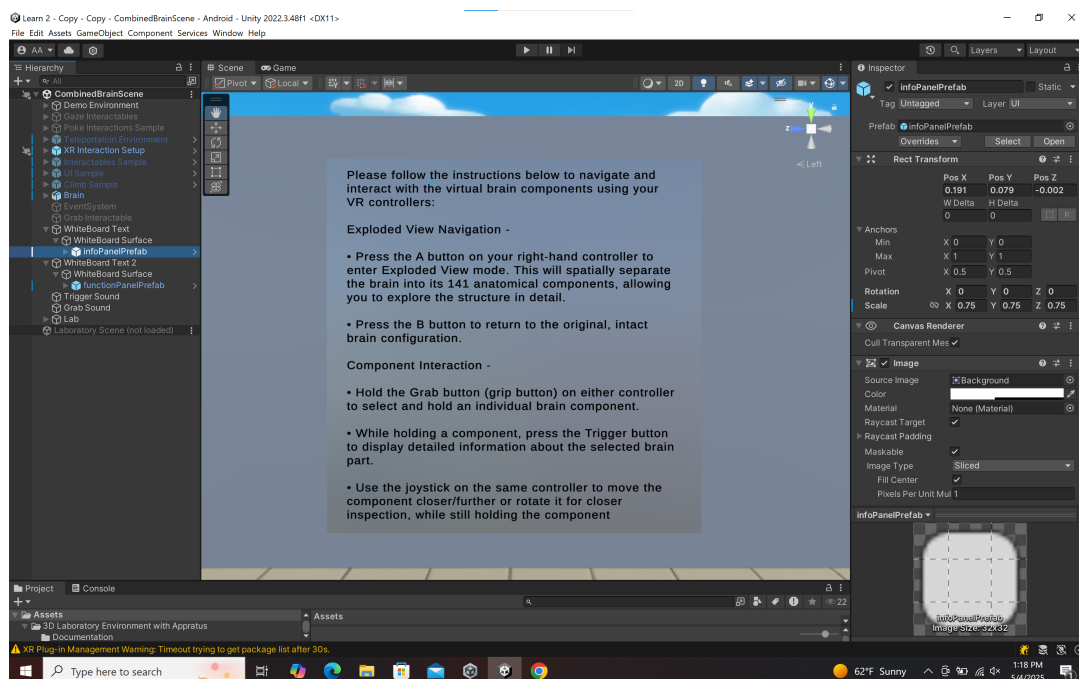


Figure 6

interacts with a brain component in VR-the script activates the info panel and function panel to make them visible, and updates the text to display the relevant information, providing users with immediate, context-specific information about the selected brain structure.

The idea of including a “associated components” section was brought up because it was not possible to highlight certain sections of the components as they were imported as 1 whole mesh, and most of the functions required overlapping mesh components to be highlighted.

5. XR Interaction - VR Deployment

Finally, the XR Interaction Setup was downloaded and imported to test out the implementation.

It helped a lot to debug a lot of the code and overall setup. I decided to turn off gravity and let the user “fly” since when the brain is “exploded” a lot of the components are exploded far off from each other, so flying helps move around freely in the space and look at each component in more details.

The instructions to move around and interact with a component are also provided on each of the whiteboards when the implementation is first loaded, and also when the components are put back together on pressing B (Figures 5 & 6).

A lab scene was also added to give the environment a more “scientific learning” effect and put the brain and whiteboards inside it. I then topped it off by adding a personalized skybox to give the environment a better and more appealing look.

User Feedback

I made a questionnaire to gather user data and get feedback on the VR implementation. The questionnaire I developed comprehensively addresses the core dimensions of usability, aesthetics, and user experience that are critical for evaluating an interactive VR educational

model. By including questions on first impressions, clarity of layout, aesthetic appeal, readability, responsiveness, consistency, ease of use, and engagement, the survey captures both the objective and subjective aspects of user interaction-ranging from visual design and navigation to how intuitive and enjoyable the system feels. The inclusion of open-ended questions for positive feedback and improvement suggestions further allows users to express nuanced opinions and specific needs, ensuring that both strengths and areas for refinement are identified. This approach aligns with best practices in VR usability research, which emphasize the importance of gathering both quantitative and qualitative data to inform iterative design improvements.

Feedback from the questionnaire highlighted desires for enhanced aesthetics and better accessibility for non-VR users. In response, I implemented two instructional whiteboards within the virtual environment to provide clear, persistent guidance, making the system more approachable for users unfamiliar with VR controls. Additionally, I created a dedicated lab scene and improved the skybox to elevate the visual atmosphere and overall immersion, directly addressing concerns about the interface's aesthetic appeal. These targeted updates demonstrate how the questionnaire not only encapsulated the overall usability of the model but also provided actionable insights that guided meaningful enhancements to the user experience.

Challenges

During the development of this VR brain model, I encountered several significant challenges, many stemming from my initial unfamiliarity with both Unity and VR development. As someone new to Unity, I had to quickly adapt to its unique component-based architecture and learn how to efficiently manage GameObjects, scripts, and physics interactions. The added complexity of

integrating the XR Interaction Toolkit for VR-specific controls and interactions introduced another layer of difficulty, needing me to understand not only how to implement VR locomotion and object manipulation but also how to optimize performance for hardware like the Meta Quest, which has strict frame rate and rendering requirements.

Another major challenge was sourcing and integrating suitable neuroanatomical data. While the Allen Brain Atlas provided a strong foundation, converting its segmented data into usable 3D models involved multiple steps, including manual cleanup and reformatting in itk-SNAP to ensure anatomical accuracy and compatibility with Unity. Gathering comprehensive hierarchical and especially functional information for each brain region also required synthesizing data from various sources, which was often time-consuming and sometimes resulted in incomplete or conflicting information that needed to be resolved.

Technical issues were frequent during the coding process. Implementing the explode and return functionality for the brain components presented difficulties in maintaining spatial relationships and avoiding unwanted mesh collisions or physics glitches. Positioning the instructional whiteboards so they remained readable and accessible in a 3D VR space proved tricky, as did ensuring that all interactive elements responded smoothly to user input. I also faced several bugs related to importing and configuring the XR toolkit, including version conflicts and deprecated methods, which required troubleshooting and sometimes reverting to earlier package versions. Finally, I had to balance the technical aspects of development with the pedagogical goals of the project. Early user feedback, gathered through a detailed questionnaire, highlighted areas for improvement such as the need for a more aesthetically pleasing environment and better guidance for new users. In response, I added two instructional whiteboards to provide persistent, clear

directions within the VR scene, and enhanced the visual atmosphere by creating a dedicated lab environment and improving the skybox. These iterative updates not only addressed specific user concerns but also made the model more accessible and engaging for a wider range of users, including those less familiar with VR. Overall, these challenges underscored the importance of adaptability, user-centered design, and continuous learning in the development of effective educational technology.

Possible Additions/Updates

The user feedback has helped me think and come up with a lot of modifications.

- A search function that helps “pop” out a brain component when searched for using a functional search bar
- A possible puzzle-mode which lets users put the brain components together in the right place to complete the brain anatomy - this is supposed to be an educational aspect which would let students learn about the anatomy of the brain in a fun interactive way.
- Another search function that lets users look for components that are responsible for a particular function.

Conclusion

In conclusion, the development of this VR brain model was not only a technical and creative endeavor but also an invaluable learning journey for me. Working on the thesis with little prior experience in Unity or VR development, I was challenged to rapidly acquire new skills in 3D modeling, interactive design, and user experience optimization. Sourcing and integrating complex neuroanatomical data, troubleshooting code bugs, and overcoming the limitations of new toolkits all pushed me well beyond my comfort zone. Each obstacle, whether it was

resolving mesh collisions in the explode/return functionality, perfecting the positioning of instructional whiteboards, or adapting to the nuances of the XR Interaction Toolkit became an opportunity for growth, deepening my understanding of both the technical and educational aspects of immersive learning environments.

Moreover, engaging with user feedback through a carefully designed questionnaire taught me the importance of iterative, user-centered design. Suggestions for improved aesthetics and accessibility prompted me to enhance the visual appeal of the environment with a new lab scene and skybox, and to add clear instructional whiteboards to guide users through the experience.

These changes not only improved the usability and engagement of the model but also reinforced the value of adaptability and responsiveness in the design process. Finally, this project has been transformative, broadening my technical expertise, sharpening my problem-solving skills, and fostering a mindset of continuous learning that I will carry into future work. The experience has shown me firsthand how embracing challenges and feedback can lead to meaningful personal and professional growth, while also contributing to innovative advancements in educational technology.

Link to GitHub:

https://github.com/akikaziz/Brain_VR

Acknowledgement:

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Resources

1. <https://community.brain-map.org/t/human-brain-atlas-mesh-files/560/2>
2. [https://atlas.brain-map.org/atlas?
atlas=138322605#atlas=138322605&plate=112360888&structure=10390&x=40320&y=469
76&zoom=-7&resolution=124.49&z=3](https://atlas.brain-map.org/atlas?atlas=138322605#atlas=138322605&plate=112360888&structure=10390&x=40320&y=46976&zoom=-7&resolution=124.49&z=3)
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4. <https://assetstore.unity.com/packages/3d/environments/chemistry-lab-items-pack-220212>